

Problem

Submissions

Leaderboard

Discussions

Given two sorted arrays, merge them and find the median of the combined sorted array.

Input Format

First line contains two integers n and m. Second line contains n sorted integers. Third line contains m sorted integers.

Constraints

$1 \leq n, m \leq 10^5$ $-10^6 \leq$ array elements $\leq 10^6$

Output Format

Print the median of the merged array.

If the total number of elements is odd, print the middle element. If even, print the average of the two middle elements with one decimal place.

Change Theme

Language

Python 3

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```
1 # Enter your code here. Read input from STDIN. Print output to STDOUT
2 n, m = map(int, input().split())
3
4 arr1 = list(map(int, input().split()))
5 arr2 = list(map(int, input().split()))
6
7 i = 0
8 j = 0
9 merged = []
10
11 while i < n and j < m:
12     if arr1[i] <= arr2[j]:
13         merged.append(arr1[i])
14         i += 1
15     else:
16         merged.append(arr2[j])
17         j += 1
18
19 while i < n:
20     merged.append(arr1[i])
21     i += 1
22
23 while j < m:
24     merged.append(arr2[j])
25     j += 1
26
27 # Find the median
28 n_m = n + m
29 if n_m % 2 == 0:
30     # Even number of elements
31     mid1 = n_m // 2 - 1
32     mid2 = n_m // 2
33     median = (merged[mid1] + merged[mid2]) / 2
34 else:
35     # Odd number of elements
36     mid = n_m // 2
37     median = merged[mid]
```

Line: 33 Col: 27

⬆️ Upload Code as File

☐ Test against custom input

Run Code

Submit Code

2 2

1 2

3 4

* Trailing spaces detected in custom input

Compilation Successful :)

Click the Submit Code button to run your code against all the test cases.

Input (stdin)

Download

1 2 2

2 1 2

3 3 4

Your Output (stdout)

1 2.5